

IT314 - Software Engineering

Lab Assignment-3 – Automated Pipeline for identifying FR and NFR.

You need to identify the stakeholders, elicitation techniques, identification of functional and non-functional requirements.

1. Background

Requirements Elicitation is one of the most human-dependent stages of the software engineering lifecycle it involves identifying stakeholders, understanding their goals, choosing the right technique to draw out information from them, and finally converting that information into Functional and Non-Functional Requirements.

In this assignment, you will build an **LLM-powered pipeline** that automates this elicitation process for a given system, using LangChain and prompting techniques covered in the previous lab.

2. Task

You are required to build a LangChain-based pipeline that takes the case study as input and progresses it through the following stages of requirements elicitation:

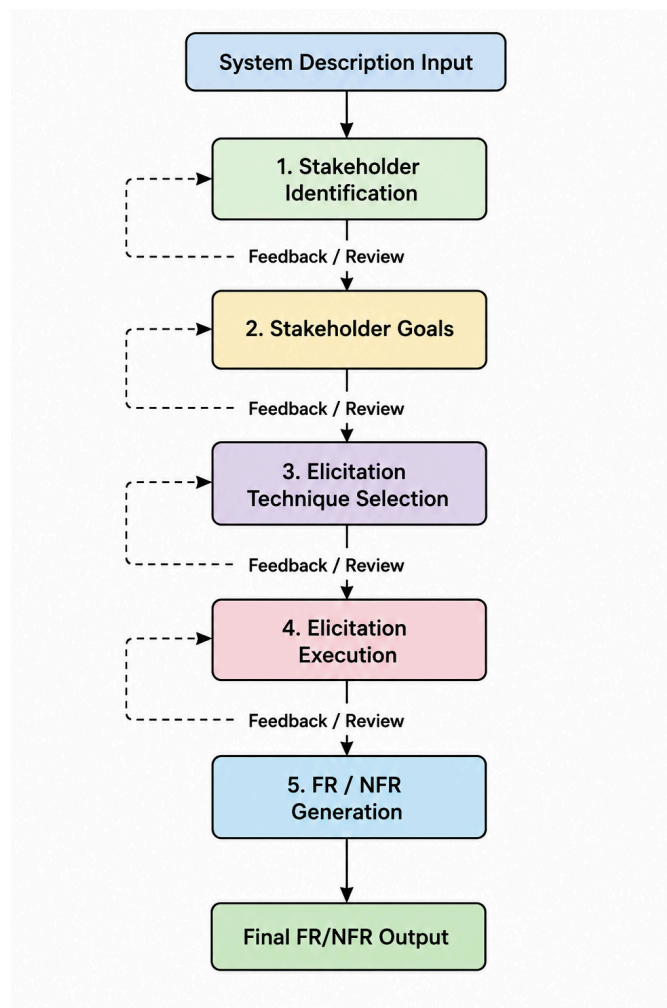
1. **Stakeholder Identification** Identify the relevant stakeholders for the given system.
2. **Stakeholder Goals** for each of the stakeholders carried forward from Stage 1, identify their goals/needs from the system.
3. **Elicitation Technique Selection** Based on the stakeholders and their goals, decide which elicitation technique(s) are appropriate (e.g., interview, questionnaire, workshop, brainstorming, observation, etc.).
4. **Elicitation Execution** Carry out the chosen technique(s) e.g., if a questionnaire is selected, generate the actual questions; if an interview is selected, decide whom to interview and draft the interview questions.
5. **FR/NFR Generation** Using everything gathered in the previous stages, generate a structured list of Functional Requirements (FRs) and Non-Functional Requirements (NFRs) for the system.

Mandatory Design Requirement Feedback at Every Stage

The pipeline **must not** run the above five stages blindly. At each stage transition, there must be **some feedback/review mechanism** that lets a review, edit, approve, or steer the output of one stage before it is carried forward into the next.

What form this feedback mechanism takes is intentionally left for you to design and justify. There is no single correct implementation think about what is practical to build and appropriate for each stage, and be ready to explain your reasoning. Hint: Human-in-the-loop (HITL), ClarifyLLM, LLM as a Judge etc., you can develop for feedback mechanism.

3. Expected Pipeline (Block Diagram)



4. Target System

Patient Information System – MentCare

A patient information system to support mental health care maintains information about patients suffering from mental health problems and the treatments they have received. Most mental health patients do not require dedicated hospital treatment but regularly attend specialist clinics. These clinics may be located in hospitals, local medical practices, or community centers to make them easier for patients to access.

The MHC-PMS (Mental Health Care-Patient Management System) is an information system intended for use in mental health clinics. It uses a centralized database of patient information but can also run on PCs at sites without secure network connectivity. When disconnected, local copies of patient records can be used and synchronized when network access is available. The system is not a complete medical records system, but it may exchange data with other clinical information systems.

The MHC-PMS has two main goals: 1. To generate management information that allows health service managers to assess performance against local and government targets. 2. To provide medical staff with timely information to support the treatment of patients.

Users include doctors, nurses, health visitors, receptionists, medical records staff, and administrative staff. The system records information about patients (name, address, age, next of kin, etc.), consultations, conditions, and treatments. It also generates reports for medical staff and health authority managers. Medical reports focus on individual patients, while management reports are anonymized and may include information about conditions, treatment costs, and clinic activity.

The key features of the system are:

1. **Individual care management:** Clinicians can create and edit patient records, view patient history, and access summaries of important problems and treatments.
2. **Patient monitoring:** The system monitors patients involved in treatment and issues warnings when possible problems are detected. It also tracks patients who have been 'sectioned' and ensures that legally required checks are carried out at the correct time.
3. **Administrative reporting:** The system generates regular management reports showing patients treated, patients entering or leaving the care system, patients sectioned, drugs prescribed, and treatment costs.

The system must comply with data protection laws governing the confidentiality of personal information and mental health laws concerning the compulsory detention of patients who may be a danger to themselves or others. Staff decisions must be recorded for possible judicial review.

Privacy and safety are critical requirements. Patient information must remain confidential and only be available to authorized medical staff and the patients themselves. Since some mental illnesses may cause patients to become suicidal or dangerous to others, the system should warn medical staff about potentially dangerous patients.

The system must also remain available when needed, as failure could compromise patient safety or prevent correct medication from being prescribed. However, there is a conflict between privacy and availability: while privacy is easier to maintain with a single copy of the data, multiple copies are required to ensure availability during server failures or when the system is disconnected from the network.

5. Deliverables

- **Working code** implementing the pipeline described above, built using LangChain and LLM API calls.
- **Prompts developed at each stage of the pipeline**
- **Output Screenshots**
- **A short report** (2–3 pages) explaining:
 - The feedback/review mechanism you designed at each stage, and why.
 - Any notable prompting or design decisions made at each stage.